

## J.A. CHARITH NIROSHAN

### Full-Stack & IoT / AI Developer

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### PROFESSIONAL SUMMARY

Tech-driven **BIT (Hons) Undergraduate (GPA: 3.39)** at the University of Vavuniya with hands-on experience in Full-Stack Web Development, Computer Vision (Python, OpenCV), and IoT Embedded Hardware Systems. Proven enterprise exposure through a Technical Internship at HIT Innovations Pvt Ltd. Seeking an Internship or Junior Software / IoT Engineering role to build impactful digital-physical solutions.

### TECHNICAL SKILLS

- **Programming Languages:** Java, Python, C#, C++ (Arduino), JavaScript (ES6+), HTML5, CSS3, SQL
- **AI & Computer Vision:** OpenCV, MediaPipe, CVZone
- **Hardware & IoT:** Arduino Microcontrollers, C++ Firmware, Smart Sensors (pH, EC, Temp), Relays
- **Tools & Frameworks:** Git, GitHub, .NET Framework (WinForms), VS Code, Visual Studio
- **Core Competencies:** Systems Analysis & Design (SAD Diagrams), Object-Oriented Programming (OOP), Software Testing & Optimization

### EDUCATION

#### Bachelor of Information Technology (BIT Hons)

Faculty of Technological Studies, University of Vavuniya | **2023 – Present** (Academic Batch 2022/2023)

- **Current Cumulative GPA: 3.39 / 4.00**
- **Key Coursework:** Web Technologies, Database Management Systems, Software Engineering & SAD, Data Structures & Algorithms, Object-Oriented Programming.

#### Diploma in Information Communication Technology

*IMBS Green Campus | Completed*

- **Key Focus:** Computing Fundamentals, Systems Architecture, Applied IT.

### **G.C.E. Advanced Level (Technology Stream)**

*Sandalankawa Central College*

- **Bio System Technology (BST):** B
- **Science for Technology (SFT):** B
- **Information & Communication Technology (ICT):** C

### **G.C.E. Ordinary Level**

*Sandalankawa Central College*

- **A Passes (3):** Religion, ICT, Geography
- **B Passes (5):** Mathematics, Science, Sinhala, History, Drama
- **C Pass (1):** English

## **WORK EXPERIENCE**

### **Technical Intern | HIT Innovations Pvt Ltd**

*Duration: 2 Months*

- Applied core software engineering, debugging, and testing practices in enterprise software projects.
- Assisted senior engineers in code optimization and agile workflows.
- Conducted system testing and documented software issue trackers.

## **KEY PROJECTS**

### **Real-Time Cricket Scoring Web App | *HTML5, CSS3, JavaScript, GitHub Pages***

- Built a responsive web app replacing manual scoresheets with real-time ball-by-ball score calculations.
- Integrated dynamic run-rate tracking algorithms and live target indicators.

### **Gesture-Controlled Snake Game | *Python, OpenCV, MediaPipe, CVZone***

- Developed a computer vision interactive game replacing keyboard inputs with real-time hand gesture tracking.

- Processed multi-point landmark detection for dynamic direction control.

#### **ANSI Color ASCII Video Player | *Python, OpenCV, ANSI Truecolor***

- Built a CLI tool that processes MP4 video streams into 24-bit Truecolor terminal ASCII art in real-time.
- Optimized frame-buffer operations to eliminate terminal rendering flickers.

#### **Smart Hydroponics Grow Tower | *Arduino C++, IoT Sensors, Actuator Relays***

- Designed an automated IoT farming system with real-time pH and EC water quality tracking.
- Programmed closed-loop firmware to dynamically trigger automated nutrient dosing pumps.

#### **Hostel Allocation System | *C#, .NET Framework, Windows Forms***

- Created a desktop administration app automating student room assignments and payment recordkeeping.
- Designed system workflows and relational database architectures using SAD principles.